Informed Search

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Artificial Intelligence

Slides adapted from materials by

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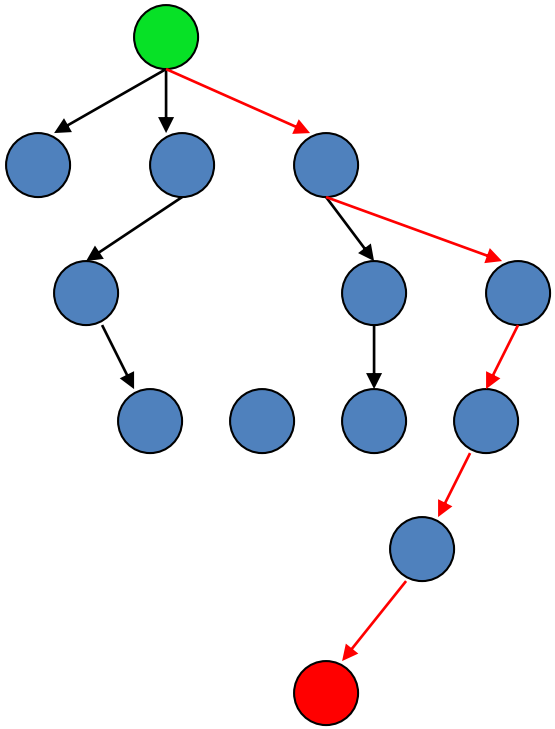
Informed (Heuristic) Search

Idea: be **smart** about what paths to try.



Blind Search vs. Informed Search

- What's the difference?

- How do we formally specify this?

  A node is selected for expansion based on an evaluation function that estimates cost to goal.

General Tree Search Paradigm

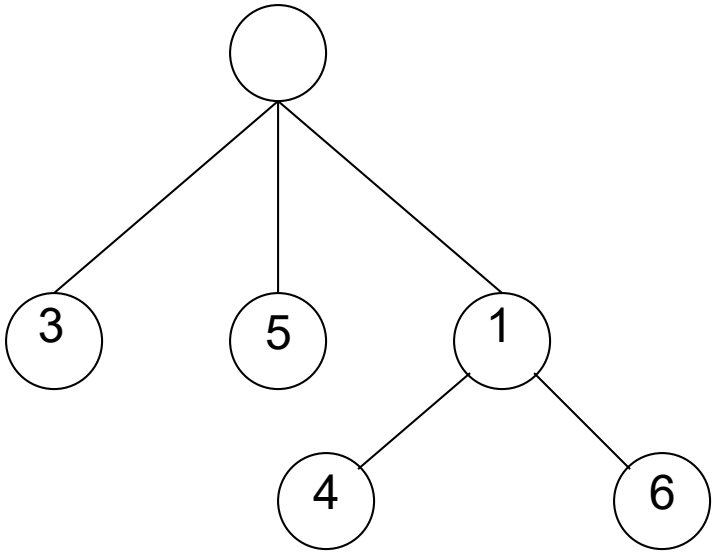
```
function tree-search(root-node)
   fringe ← successors(root-node)
   while ( notempty(fringe) )
         {node ← remove-first(fringe) //lowest f value
            state ← state(node)
            if goal-test(state) return solution(node)
            fringe ← insert-all(successors(node),fringe) }
   return failure
end tree-search
```

General Graph Search Paradigm

```
function tree-search(root-node)
   fringe ← successors(root-node)
   explored ← empty
   while ( notempty(fringe) )
         {node ← remove-first(fringe)
             state ← state(node)
             if goal-test(state) return solution(node)
             explored ← insert(node,explored)
             fringe ← insert-all(successors(node),fringe, if node not in explored)
         }
   return failure
end tree-search
```

Best-First Search

- Use an evaluation function f(n) for node n.
- Always choose the node from fringe that has the lowest f value.

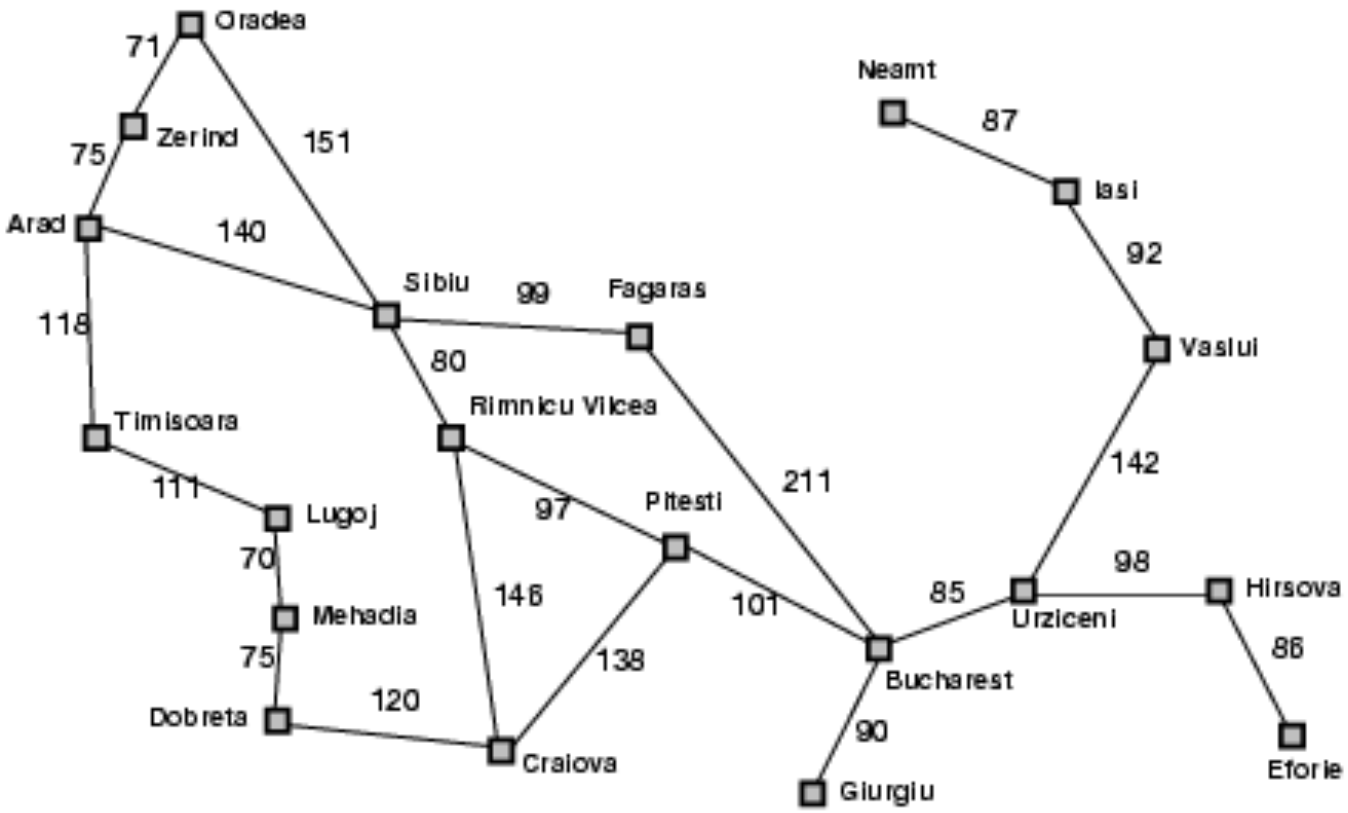


Best-first search

- A search strategy is defined by picking the order of node expansion
- Idea: use an evaluation function *f(n)* for each node
  - estimate of "desirability“

  → Expand most desirable unexpanded node
- Implementation:
  Order the nodes in fringe in decreasing order of desirability
- Special cases:
  - greedy best-first search
  - A* search

Romania with step costs in km



Old (Uninformed) Friends

- Breadth First =
  - Best First
  - with f(n) = depth(n)

- Uniform cost search =
  - Best First
  - with f(n) = the sum of edge costs from start to n

g(n)

Greedy best-first search

- Evaluation function $f(n) = h(n)$ (heuristic function)
  = estimate of cost from n to *goal*

- e.g., $h_{SLD}(n)$ = straight-line distance from n to Bucharest

- Greedy best-first search expands the node that appears to be closest to goal

Properties of greedy best-first search

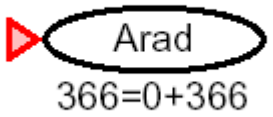
- Complete?
  - No – can get stuck in loops, e.g., Iasi → Neamt → Iasi → Neamt →
- Time?
  - $O(b^m)$, but a good heuristic can give dramatic improvement
- Space?
  - $O(b^m)$ -- keeps all nodes in memory
- Optimal?
  - No

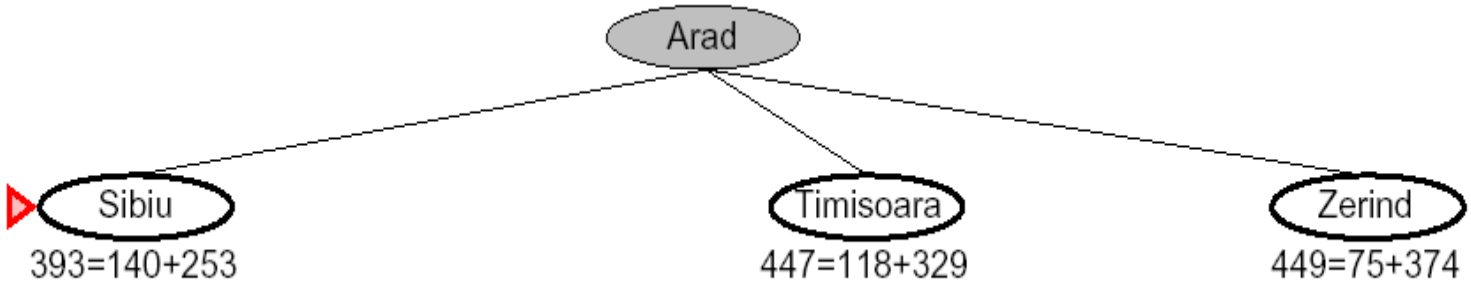
A* search

- Idea: avoid expanding paths that are already expensive
- Evaluation function $f(n) = g(n) + h(n)$

- $g(n)$ = cost so far to reach n
- $h(n)$ = estimated cost from n to goal
- $f(n)$ = estimated total cost of path through n to goal

A* for Romanian Shortest Path

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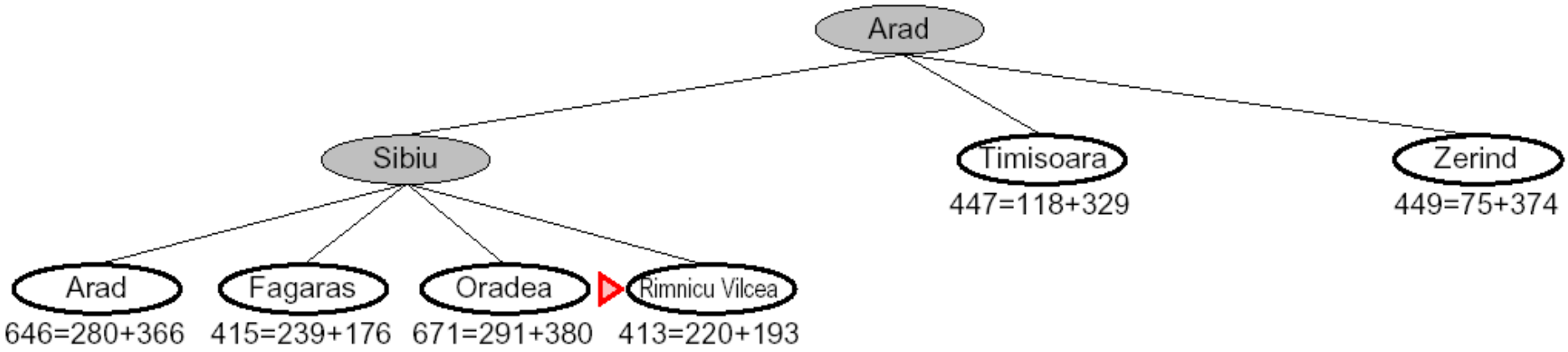
Timisoara
447=118+329
Zerind
449=75+374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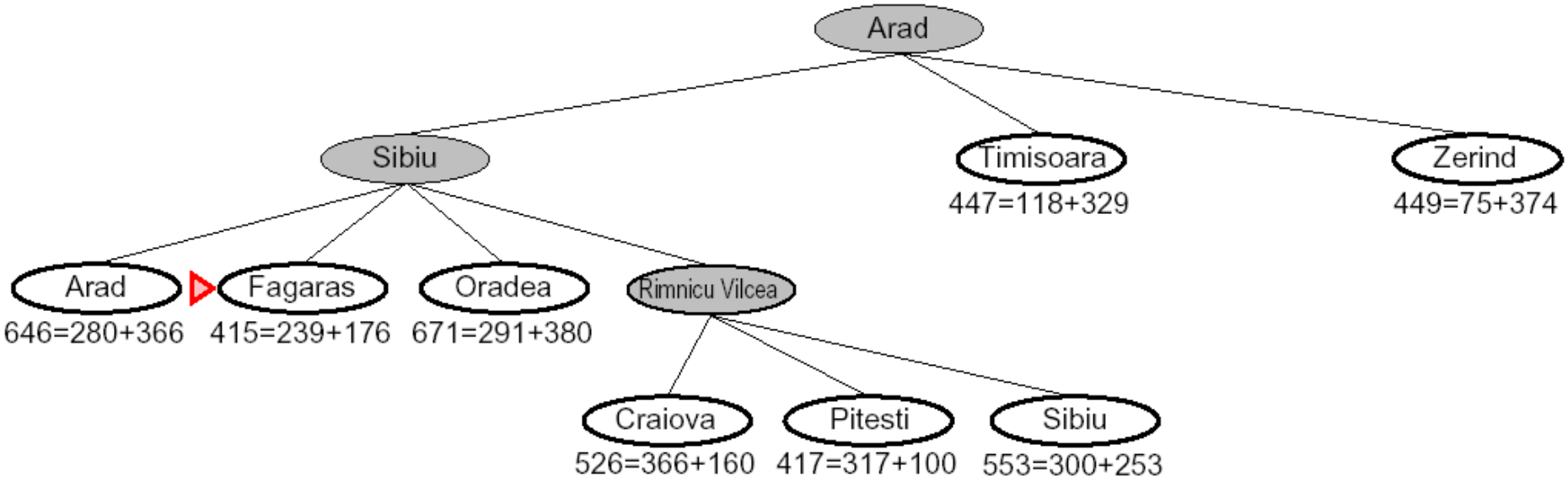
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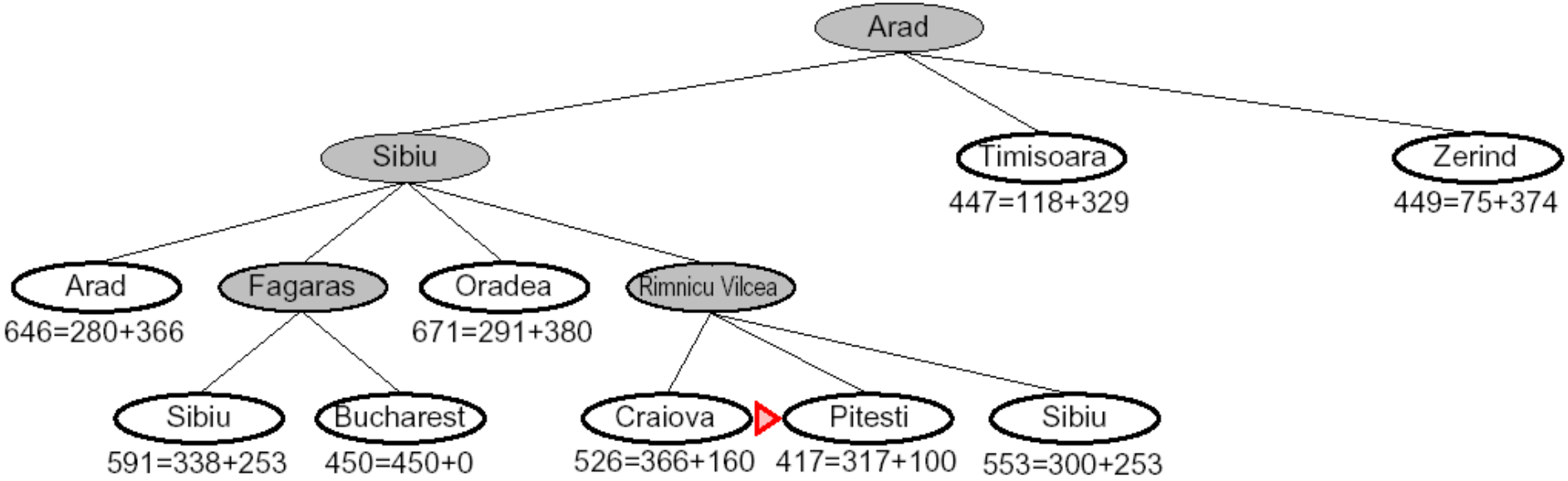
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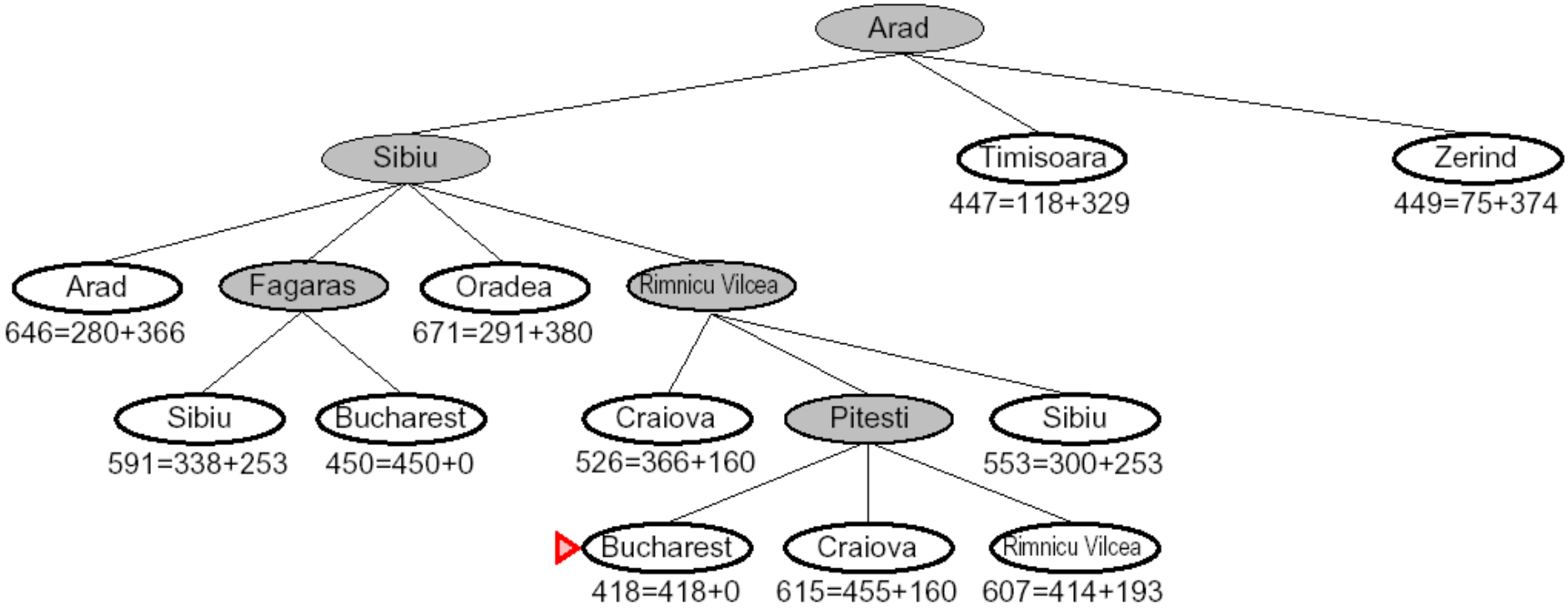
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Admissible heuristics

- A heuristic function $h(n)$ is admissible if for every node n, $h(n) \leq h^*(n)$, where $h^*(n)$ is the true cost to reach the goal state from n.

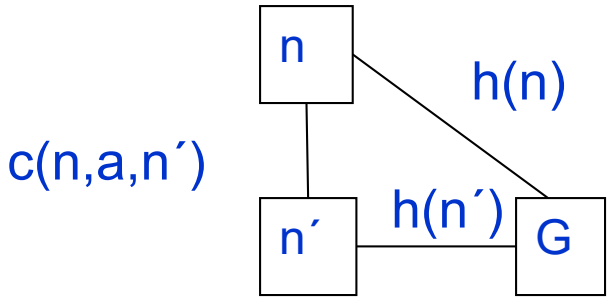
- An admissible heuristic never overestimates the cost to reach the goal, i.e., it is optimistic

- Example: $h_{SLD}(n)$ (never overestimates the actual road distance)

- Theorem: If $h(n)$ is admissible, A* using `TREE-SEARCH` is optimal

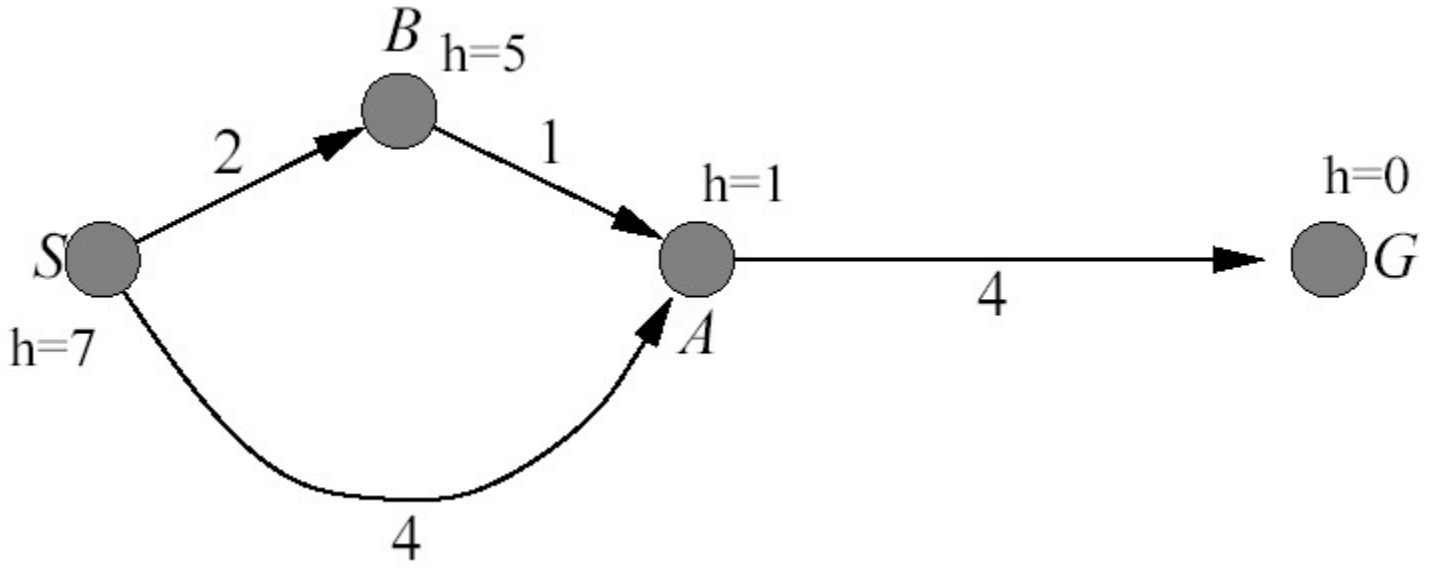
Consistent Heuristics

- h(n) is consistent if
  - for every node n
  - for every successor n´ due to legal action a
  - h(n) <= c(n,a,n´) + h(n´)



- Every consistent heuristic is also admissible.
- Theorem: If *h(n)* is consistent, A* using GRAPH–SEARCH is optimal

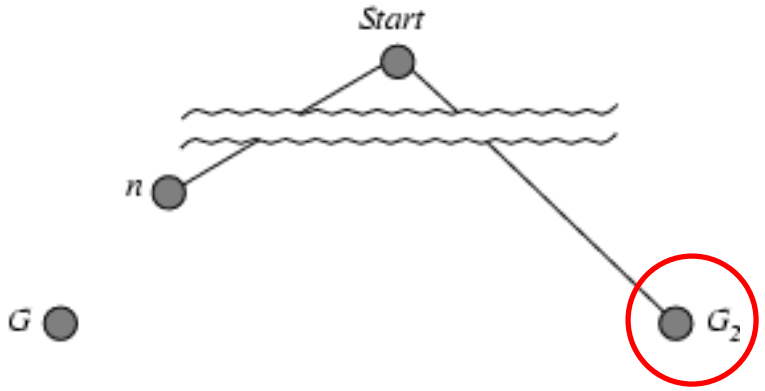
Example



Source: http://stackoverflow.com/questions/25823391/suboptimal-solution-given-by-a-search

Proof of Optimality of (Tree) A*

- Assume *h*() is admissible.
  Say some sub-optimal goal state G_2 has been generated and is on the frontier.
  Let *n* be an unexpanded state such that *n* is on an optimal path to the optimal goal *G*.



Focus on G_2:

$f(G_2) = g(G_2)$ since $h(G_2) = 0$

$g(G_2) > g(G)$ since G_2 is suboptimal

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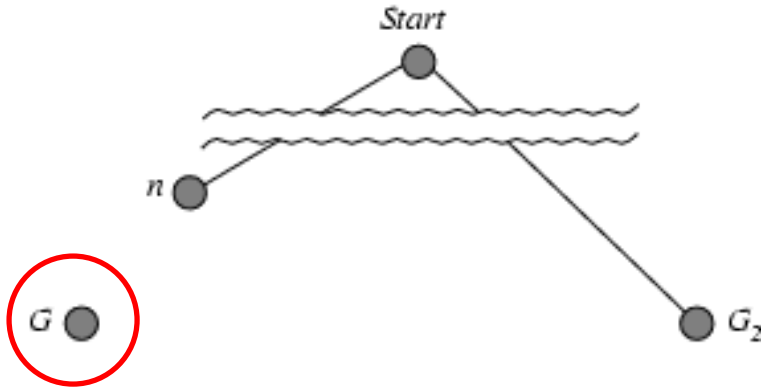

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$g(G_2) > g(G)$ since G_2 is suboptimal

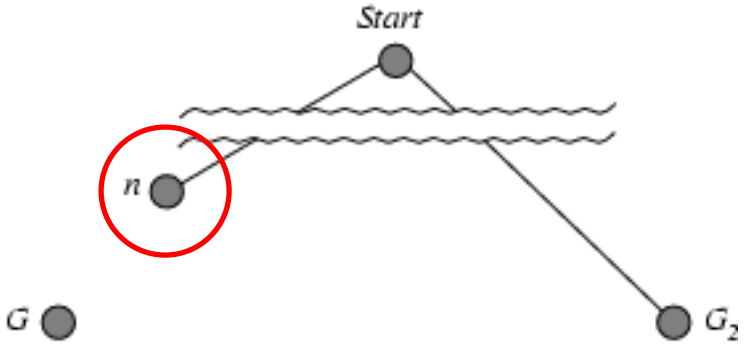
Focus on G:

$f(G) = g(G)$ since $h(G) = 0$

$f(G_2) > f(G)$ substitution

Proof of Optimality of (Tree)A*

- Assume *h*() is admissible.
  Say some sub-optimal goal state G_2 has been generated and is on the frontier.
  Let *n* be an unexpanded state such that *n* is on an optimal path to the optimal goal *G*.



$f(G_2) = g(G_2)$ since $h(G_2) = 0$
$g(G_2) > g(G)$ since G_2 is suboptimal

$f(G) = g(G)$ since $h(G) = 0$
$f(G_2) > f(G)$ substitution

Now focus on n:

$h(n) \le h^*(n)$ since h is admissible
$g(n) + h(n) \le g(n) + h^*(n)$ algebra
$f(n) = g(n) + h(n)$ definition
$f(G) = g(n) + h^*(n)$ by assumption
$f(n) \le f(G)$ substitution

Hence $f(G_2) > f(n)$, and A^* will never select G_2 for expansion.

Properties of A*

- Complete?

  Yes (unless there are infinitely many nodes with f $\leq f(G)$)

- Time? Exponential (worst case all nodes are added)

- Space? Keeps all nodes in memory

- Optimal?

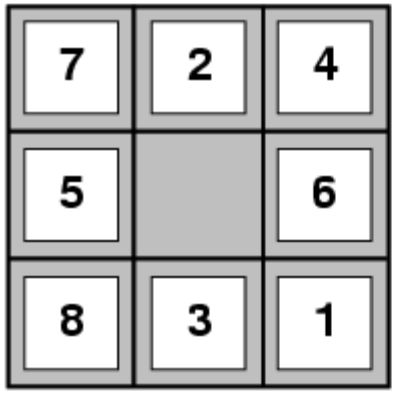
  Yes (depending upon search algo and heuristic property)

Admissible heuristics

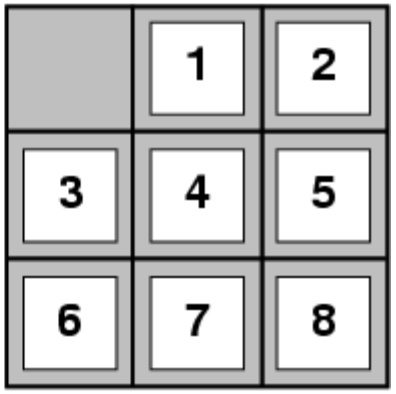
E.g., for the 8-puzzle:

- $h_1(n)$ = number of misplaced tiles
- $h_2(n)$ = total Manhattan distance

(i.e., no. of squares from desired location of each tile)



Start State



Goal State

- $h_1(S) = ?$
- $h_2(S) = ?$

Admissible heuristics

E.g., for the 8-puzzle:

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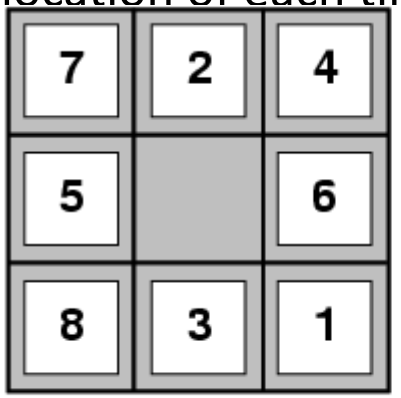

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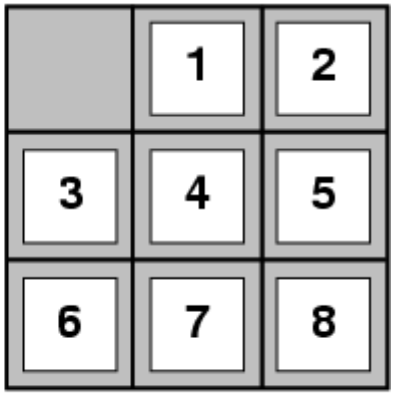

Goal State

- $h_1(S) = ?$ 8
- $h_2(S) = ?$ 3+1+2+2+2+3+3+2 = 18